Ulster University

Intelligent Systems Research Centre

Computational Neuroscience, Neurotechnology and Neuro-inspired Artificial Intelligence (ISRC-CN³)

ISRC-CN³

Summer School

12 – 19 August 2026

Intelligent Systems Research Centre

Ulster University Derry~Londonderry

Information Booklet

Welcome Note

There have been rapid advancements and investments in research and development in brain sciences, neurotechnology, neural data modelling and neuro-inspired artificial intelligence (AI). These advancements have not only led to a deeper understanding of brain functions and disorders, but also the development and application of powerful AI and machine-learning algorithms that affect our everyday lives. In fact, historically, AI was inspired by how intelligence arises from the brain.

The Computational Neuroscience, Neurotechnology and Neuro-inspired AI (CN^3) Summer School (https://www.ulster.ac.uk/conference/isrc-cn3-summer-school) aims to train the next generation of researchers on these state-of-the-art developments. This short course will touch on the areas of computational neuroscience, neural data science and signal processing, neurotechnology and neuro-inspired AI. The School is unique in that it offers important and timely topics that are not covered in other Schools or taught courses, or are delivered only individually, in an integrated way, ranging from pedagogical to advanced levels. Although neural computation and neuro-inspired AI research are conducted in the island of Ireland, there is very little relevant training and taught courses, especially for early career researchers, in the region; this School aims to bridge this gap.

On this note, the organising committee warmly welcome you to attend the Summer School!

About the Intelligent Systems Research Centre

The Summer School will be held at the Intelligent Systems Research Centre (ISRC: https://www.ulster.ac.uk/research/topic/computer-science/intelligent-systems-research-centre), a major research unit within the School of Computing, Engineering and Intelligent Systems at Ulster University in Derry ∼ Londonderry, Northern Ireland. This is the sixth ISRC-CN^3 Summer School. The ISRC is dedicated to developing a bio-inspired computational basis for AI to power future AI technologies. This is achieved through understanding how the brain works at multiple levels, from cells to cognition, and applying that understanding to create models and technologies that solve complex issues that face people and society. To accomplish this, a variety of research strategies and applications are used, including big data and machine learning, brain imaging and neural interfacing, human-computer interaction and neuromorphic computing.

The ISRC is housed in a large, purpose-built facility, with state-of-the-art resources, including neuroimaging, neurotechnology and robotic facilities, and a high-performance computing (HPC) facility for big data analytics and large-scale computational simulations. There will be a tour of labs for in-person attendees. The ISRC is multidisciplinary, with arguably the largest cluster of computational neuroscientists and neuro-inspired AI researchers in the island of Ireland, with strong collaborations with many clinical, biomedical, neuroscience, AI and mental health centres, and industrial partners, allowing its research output to quickly translate into applications.

Summer School Structure

This booklet and pre-school materials, including mathematical and programming notes, have been made available in advance of the event for attendees to review (GitHub links will be sent by separate e-mail).

Required software (Python and MATLAB) can be downloaded and configured before the event.

Web links to attend the lecture and lab sessions will be sent to all applicants closer to the dates of the summer school. Hence, please check your email (and your junk folder) regularly. Information on joining the guest wi-fi accounts will be provided on Day 1 for online access to materials while on campus.

Unless mentioned for a last-minute change, the lecture room will be located in rooms **MU308** within the **MU** building. Computer lab sessions will be held in the **Timberquay building** of the campus. The tutors will guide the participants to the lab in the first day of the School.

Online attendees are themselves responsible for accessing reliable internet. When not speaking online, please remember to turn off the microphone and video camera to avoid echo effects and hanging up during video streaming. At the end of the lecture/talk, for questions and answers, you may turn on your microphone and video camera to ask questions or speak to the lecturer/speaker. You can also ask through the chat platform. During lab sessions, you can ask Tutors questions either verbally or through chat. But please be mindful that we have limited Tutors per lab session. Both lecture sessions and lab sessions will be recorded for attendees' viewing.

Towards the end of the Summer School, feedback from attendees will be requested. Anonymity of feedback is optional. This will be used in reviews and reporting, and for improving future versions of the Summer School. **A certificate of participation will also be provided upon completion of the Summer School.**

Lectures:

Lectures, including external speakers, will be delivered throughout the day with several breaks within this period. The schedule of the lecture can be found in the website.

Attendees are encouraged to attend as many of the lectures as possible, as the content of the presentations may be built on that of previous presentations. **Lectures will be delivered both physically and online (live streaming)**. Physical lectures will be broadcast live to those attending the fully online version. Participants attending online will be able to ask questions using their computer's microphone and webcam or by submitting them through the chat function on the online platform. Unless otherwise requested by the speaker, all lectures will be recorded and made available after the event. This will enable participants who are unable to attend live (e.g. due to time zone differences, work commitments, or other personal responsibilities) to view the sessions at a later time, as well as allowing attendees to revisit the material. Information about the video clips will be made available through our GitHub repository (links will be sent out in a separate email).

Lecture Room:

- **Room MU308 - MU Building**

Labs:

Each computer lab session aims to consolidate and reinforce the topics delivered during the lectures of that day. Lab sessions will take place in **the Timberquay building** from 5:15 PM, and they will be led by Tutors. Attendees are highly encouraged to explore additional aspects as an independent study.

Computer labs will be conducted in Python and MATLAB. If MATLAB is not available, students joining online can download MATLAB's 30-day free trial version or MATLAB's online version (https://uk.mathworks.com/products/matlab-online.html).

Codes will be provided by the Tutors and available on GitHub (links to be provided a day before). Data will be provided when needed. We recommend that attendees, especially online attendees, download the relevant software to their own personal computer before the Summer School. Attendees with limited mathematics and computer programming experience should review the prepared mathematical notes or other resources, such as https://www.datacamp.com/, before attending the Summer School.

In-person attendees are recommended to attend as many of these lab sessions as possible. **It is optional for online attendees to attend the lab sessions, and they can work on the lab materials independently.** Tutors will be available through Teams chat and Slack to help with the Lab. Those who are attending physically will be able to access our computer lab's machines and other computing facilities. Guest accounts will be provided for all in-person attendees.

Lab sessions will be partially recorded (especially at the beginning and during demonstrations) to allow those who were unable to attend (e.g. due to time zone differences, work-related or other personal responsibilities) or for revisits. We will provide the information on the video clips on our GitHub links.

Computer Lab:

- Timberquay building

ISRC-CN^3 GitHub link:

Notes, codes, datasets, and video clips will be made available at our ISRC-CN^3 GitHub link https://github.com/ISRC-CN3.

For those who are not familiar with computer programming or mathematics, it is advisable that they read, refresh or practise the provided materials (see Reading Materials repository in GitHub) before the start of the Summer School.

Reimbursements, claims and refunds:

If you are seeking (e.g. travel) reimbursements and claims, or refunds, please remember to save hard copies of your receipts. Then contact Dr Saugat Bhattacharyya for a claim form to be filled out.

Please note that in-person attendees who are attending only for a few days will still be paying the full fee.

Social Activities

Breakfast, lunch, and refreshments (tea and coffee) will be provided each day throughout the Summer School.

A guided walking tour of the city has been organised for 15 August 2026. In addition, a Gala Dinner will be held on Day 3 (14 August 2026) at Stitch Weave, located in Ebrington Square.

Communication

All participants will be invited to join the ISRC-CN^3 Slack channel: https://join.slack.com/t/isrc-cn32026/shared_invite/zt-44dxpx99c-STLlSfnMIdHVEphX_lLfAw

In-person attendees may use the Slack platform for planning shared accommodation.

Certification:

Certificate of Attendance will be emailed to all attendees after the end of the Summer School.

Organising committee and contacts:

- Dr. Saugat Bhattacharyya (Chair) (s.bhattacharyya@ulster.ac.uk)
- Dr. Cian O'Donnell (Co-chair) (c.odonnell2@ulster.ac.uk)
- Dr. Barry Dillon (Scientific) (B.Dillon@ulster.ac.uk)
- Prof. KongFatt Wong-Lin (Scientific) (k.wong-lin@ulster.ac.uk)
- Dr. Bronac Flanagan (Scientific) (b.flanagan@ulster.ac.uk)
- Dr. Muskaan Singh (Scientific) (M.Singh@ulster.ac.uk)

- Louise Gallagher (Secretary, Treasurer)
- Michelle Stewart (Secretary)
- Elaine Duffy (Secretary)
- Gerarld Hasson (Secretary)
- Cheryl Mullan (Secretary)
- Eoghan Tucker (Secretary)

ISRC-CN^3 logo design:

- Niall McShane
- KongFatt Wong-Lin

Web design and development:

- Dorothy McIlroy
- Cheryl Mullan
- Mark Millar
- Roger James
- Saugat Bhattacharyya

IT Support

- Christopher Hasson
- Chris O'Connell

DAY 1 (12TH AUGUST 2026, WEDNESDAY)

Morning Session

08:30 - 09:00 (BST)	Registration and Breakfast at MU building, Room Number 309's lobby
09:00 - 09:15 (BST)	Welcome Address – *Saugat Bhattacharyya*
09:15 - 10:45 (BST)	Neuroanatomical Basis of Neuroimaging-related Computational Modelling - *Prof. Dara Cannon*
11:00 - 12:30 (BST)	Modelling astrocyte-neuron interactions and glial cells – *Prof. Liam McDaid*

Afternoon Session

12:30 - 14:30 (BST)	Lunch and Campus Tour
14:30 - 16:30 (BST)	Part 1: Philosophy of Computational Modelling. Part 2: Computational Modelling of Synaptic Plasticity and Learning in Brains – *Dr. Cian O'Donnell*
16:30 - 17:00 (BST)	Tutor Interaction Session
17:15 - 19:00 (BST)	Lab session 1 – Fundamentals of Python & MATLAB programming (notes provided in advance) – *Dr. Sanjay Dutta and Mohammad Faraz*

DAY 2 (13TH AUGUST 2026, THURSDAY)

Morning Session

08:30 - 09:00 (BST)	Breakfast at MU building, Room Number 308's lobby
09:00 - 11:00 (BST)	Mathematics for Neuroscience – *Dr. Barry Dillon*
11:15 - 13:15 (BST)	Introduction to Neuronal Modelling – *Dr. Cian O'Donnell and Prof. KongFatt Wong-Lin*

Afternoon Session

13:15 - 15:00 (BST)	Lunch and ISRC Tour
15:00 - 17:00 (BST)	Modelling the dynamics of decision-making – *Prof. KongFatt Wong-Lin*
17:15 - 19:00 (BST)	Lab session 2 – Modelling neurons, neural networks & cognition – *Mohammad Faraz and Emmet Thompson*

DAY 3 (14TH AUGUST 2026, FRIDAY)

Morning Session

08:30 - 09:00 (BST)	Breakfast at MU building, Room Number 308's lobby
09:00 - 11:00 (BST)	Neural Signal Processing and Neurorehabilitation – *Prof. Girijesh Prasad*
11:15 - 12:45 (BST)	Fundamentals of functional and effective connectivity and their applications to neurodevelopmental and mental health conditions - *Dr. Maria Dauvermann*

Afternoon Session

12:45 - 15:00 (BST)	Lunch and NIFBM Facility Demo
15:00 - 16:15 (BST)	Brain Connectivity in Bipolar Disorder - *Prof. Dara Cannon*
16:15 - 17:10 (BST)	Panel Discussion - Prof. Dara Cannon, Dr. Cian O'Donnell and Dr. Maria Dauvermann
17:30 - 19:00 (BST)	Lab session 3 – Neural Signal Processing – *Mark Butler & Monika Sharma*
19:30 - (BST)	Social activity - Gala Dinner at Stitch and Weave, Ebrington Square

DAY 4 and 5 (15TH and 16TH AUGUST 2026)

Social Activity and Project Work

Saturday - Derry∼Londonderry walking tour

DAY 6 (17TH AUGUST 2026, MONDAY)

Morning Session

08:30 - 09:00 (BST)	Breakfast at MU building, Room Number 308's lobby
09:00 - 10:30 (BST)	Characterising Sensorimotor Representations with Multivariate Pattern Analysis – *Dr. Ingrid Odermatt*
10:45 - 12:15 (BST)	Brain-computer Interfacing Technology to improve User Engagement and Decision Making – *Dr. Saugat Bhattacharyya*

Afternoon Session

12:15 - 15:00 (BST)	Lunch and Networking (provided at MU building)
15:00 - 17:00 (BST)	AI-enabled wearable neurotechnology for communication and rehabilitation – *Prof. Damien Coyle*
17:15 - 19:00 (BST)	Lab session 4 - Neural Engineering – *Mark Butler & Monika Sharma*

DAY 7 (18TH AUGUST 2026, TUESDAY)

Morning Session

09:00 - 09:30 (BST)	Breakfast at MU building, Room Number 308's lobby
09:30 - 11:00 (BST)	Brain-inspired Control Architectures for Robotics and AI – *Prof. Tony Prescott*
11:15 - 12:45 (BST)	Neuromorphic Systems – *Prof. Elisa Donati*

Afternoon Session

12:45 - 14:15 (BST)	Lunch and Networking (provided at MU building)
14:15 - 15:45 (BST)	Introduction to neuromodulation – using translational models to understand and improve neuromodulation therapies – *Dr. Judith Evers*
16:15 - 17:00 (BST)	Panel Discussion - Prof. Tony Prescott, Prof. KongFatt Wong-Lin and Dr. Elisa Donati
17:15 - 19:00 (BST)	Independent Project Activity & Student Presentation Preparation

DAY 8 (19TH AUGUST 2026, WEDNESDAY)

Morning Session

09:00 - 09:30 (BST)	Breakfast at MU building, Room Number 308's lobby
09:30 - 12:30 (BST)	Student Presentation

Afternoon Session

12:30 - 14:00 (BST)	Lunch and Networking (provided at MS building)
14:00 - 15:30 (BST)	Stakeholder Talk from Bitbrain and IEEE SMC
15:30 - 15:45 (BST)	Prize-giving ceremony and closing remarks

Profiles of Speakers

Prof. Dara Cannon

Bio: Dara Cannon is a Professor of Neuroscience at the University of Galway on the West Coast of Ireland. Her research laboratory focuses on the biological underpinnings of psychiatric disorders, including analyses using brain imaging, genetics, cognitive measures, and life context, with a teaching focus on gross anatomy and magnetic resonance imaging-based neuroanatomy and neuroimaging.

Lecture Title: Neuroanatomical Basis of Neuroimaging-related Computational Modelling

Date Time: 12th August 2026, 9:15am BST
Meeting Link: https://teams.microsoft.com/meet/373366101845831?p=NkBw1nIx1wS6HEzoly

Lecture Title: Brain Connectivity in Bipolar Disorder

Date Time: 14th August 2026, 3:00pm BST
Meeting Link: https://teams.microsoft.com/meet/327948291256108?p=crGoc1wcq8jnOh6BRt

Prof. Liam McDaid

Bio: Liam McDaid is Professor of Computational Neuroscience at Ulster University and group leader for the CNET research group. His research primarily focuses on mathematical models for neuronal and astrocyte cells. Prof McDaid has successfully led projects in this area and received funding from local and international funding bodies (EPSRC, MRC, DSTL, eFutures, HFSP and HEA). Highlights of his work include the development of mathematical models on bi-directional coupling between neuronal and astrocyte cells, and models that capture: Glutamate (Glu), Potassium (K+) and Sodium (Na+) uptake by astrocytes; IP3 induced periodic release of calcium (Ca2+) from intracellular stores; and synaptic regulation by the endocannabinoid system. He has supervised around 20 PhD students and has over 200 publications.

Lecture Title: Modelling astrocyte-neuron interactions and glial cells

Date Time: 12th August 2026, 11:00am BST
Meeting Link: https://teams.microsoft.com/meet/360778520094133?p=DkRAXF0zvSWbNP7EBs

Dr. Cian O'Donnell

Bio: Cian O'Donnell did a B.Sc. in Applied Physics at Dublin City University, followed by an M.Sc. and Ph.D. in Neuroinformatics at University of Edinburgh where he studied biophysical models of electrical noise and synaptic plasticity in single neurons. He then worked for 3 years as a postdoc in the Salk Institute in La Jolla, California modelling synaptic plasticity in neural circuits, and analysing neural population activity data from mouse models of autism. From 2015-2021 he was a lecturer at the University of Bristol, then in October 2021 he joined Ulster University at Magee as a Lecturer in Data Analytics. His research group has 3 postdoctoral RAs and 6 PhD researchers, working on three topics: 1) learning and memory in the brain; 2) neural circuit dysfunction in autism; 3) statistical methods for neuroscience data.

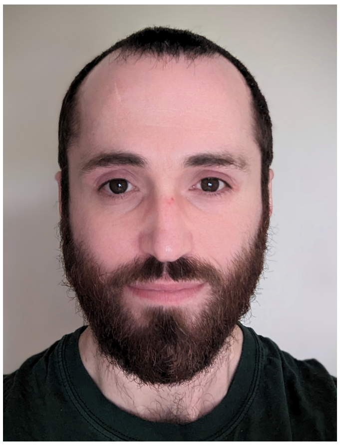
Website here: https://odonnellgroup.github.io.

Lecture Title: Part 1: Philosophy of Computational Modelling; Part 2: Computational Modelling of Synaptic Plasticity and Learning in Brains

Date Time: 12th August 2026, 2:30pm BST
Meeting Link: https://teams.microsoft.com/meet/393962323295337?p=JISGFVPqgLG27y0Cn4

Lecture Title: Introduction to Neuronal Modelling (with Prof. KongFatt Wong-Lin)

Date Time: 13th August 2026, 11:15am BST
Meeting Link: https://teams.microsoft.com/meet/313508015583879?p=YmGI6bSSmnqalPJDem



Dr. Barry Dillon

Bio: In 2016, Barry completed his PhD in theoretical physics, where he studied the phenomenology of beyond the standard model physics scenarios. He then held three postdoctoral research positions at the University of Plymouth (2017-2018), Jozef Stefan Institute (2018-2020), and the University of Heidelberg (2020-2023). Since 2018, his research has focused on applying machine-learning tools to particle physics phenomenology, particularly in the search for new physics at the Large Hadron Collider. In 2023, he worked in machine-learning research and development at AllstateNI, before taking up a position at Ulster University this year as a Lecturer in Mathematics.

Lecture Title: Mathematics for neuroscience - An introduction

Date & Time: 13th August 2026, 9:00am BST
Meeting Link: https://teams.microsoft.com/meet/343280921685856?p=nkz2WR3ugD5LrNWjR5

KongFatt Wong-Lin

Bio: KongFatt Wong-Lin is a Professor of Computational Neuroscience and Machine Intelligence. He leads the Cognitive Neuroscience and Neurotechnology research team at the Intelligent Systems Research Centre in Ulster University. His research interests span computational modelling and mathematical analysis in systems and cognitive neuroscience, psychology, brain disorders, neural computation and engineering, AI, and data science. He received his Ph.D. in Physics with a focus on Computational Neuroscience at Brandeis University, with affiliation to the Volen National Centre for Complex Systems. He was later a research associate at Princeton University, with affiliation to The Program in Applied and Computational Mathematics, and Princeton Neuroscience Institute. He has also received visiting fellowships at the University of Galway and the University of Oxford.

Website: https://www.ulster.ac.uk/staff/k-wong-lin

Lecture Title: Introduction to Neuronal Modelling (with Dr. Cian O'Donnell)

Date Time: 13th August 2026, 11:15am BST
Meeting Link: https://teams.microsoft.com/meet/313508015583879?p=YmGI6bSSmnqalPJDem

Lecture Title: Modelling the dynamics of decision-making

Date & Time: 13th August 2026, 3:00pm BST
Meeting Link: https://teams.microsoft.com/meet/329457683019209?p=rufzyGJWxXtJiHNESF

Prof. Girijesh Prasad

Bio: Prof. Girijesh Prasad is Professor of Intelligent Systems in the School of Computing, Engineering and Intelligent Systems, Ulster University (UU), UK. He is Director of Northern Ireland Functional Brain Mapping (NIFBM) facility at UU's Intelligent Systems Research Centre, where he leads the Cognitive Neuroscience and Neurotechnology research team.

He received a BTech in Electrical Engineering from Regional Engineering College (now National Institute of Technology) Calicut, India in 1987, an MTech in Computer Science and Technology from University of Roorkee (now Indian Institute of Technology Roorkee), India in 1992, and a PhD in Electrical Engineering from Queen's University of Belfast, UK in 1997. He is a Chartered Engineer, a Fellow of IET, a Fellow of Higher Education Academy, a Senior Member of IEEE, and a founder member of IEEE Systems, Man, and Cybernetics society's Technical Committee on Brain-Machine Interface Systems. In 2017, he was awarded the Fellowship of International Academy of Physical Sciences (IAPS) India, and the Senior Distinguished Research Fellowship of Ulster University. Prof. Prasad joined Ulster University, as a Lecturer in 1999; he was promoted to Senior Lecturer in 2007, Reader in 2008, and Professor in 2011. Previously he worked in industry first as a Digital Systems Engineer and then as a Power Plant Engineer in India, and as a Research Fellow on an EPSRC/industry project at Queen's University of Belfast, UK.

His research interests are in intelligent systems, data engineering, brain modelling, brain-computer interface (BCI) & neuro-rehabilitation, and assistive technology. Under his supervision, an advanced rehabilitation protocol has been developed incorporating an active physical practice stage followed by a mental practice stage, using a neuro-rehab system consisting of a robotic hand exoskeleton and an EEG/EEG-EMG based BCI, which has been trialled on groups of chronic stroke patients in UK as well as India, resulting in transformative change in patients' quality of life. He has published over 285 research papers in journals, edited books, and conference proceedings. He has supervised to completion 22 PhD students. His research has attracted 18 research grant awards amounting to over £10M funding from national and international agencies including Invest Northern Ireland, Department of Employment and Learning, Research Councils UK (RCUK), Leverhulme Trust, Royal Society, UK India Education and Research Initiative (UKIERI), UK Research and Innovation (UKRI) and Irish industry.

Websites: https://pure.ulster.ac.uk/en/persons/girijesh-prasad ; https://scholar.google.com/citations?view_op=list_works&hl=en&hl=en&user=xPw66a0AAAAJ

Lecture Title: Neural Signal Processing and Neuro-rehabilitation

Date & Time: 14th August 2026, 9:00am BST

Meeting Link: https://teams.microsoft.com/meet/326675434518608?p=1XZabsQa5D2gEwrsHY

Dr. Maria Dauvermann

Bio: Dr Dauvermann's research focuses on the identification of risk and resilience markers in young people who are at high risk of developing neurodevelopmental and mental health conditions, and is also interested in the characterisation of biopsychosocial prognostic markers of clinical and functional outcome. She uses cognitive neuroscientific and interdisciplinary methods to integrate neurobiological, psychological and psychosocial factors to better understand how youth vulnerability can influence and be influenced by neurodevelopmental and mental health conditions.

Lecture Title: Fundamentals of functional and effective connectivity and their applications to neurodevelopmental and mental health conditions

Date & Time: 14th August 2026, 11:15am BST
Meeting Link: https://teams.microsoft.com/meet/391113405983654?p=f7rYtjdQ5FvWHkL7tQ

Dr. Ingrid Odermatt

Bio: **Ingrid Odermatt** received her Bachelor's and Master's degrees in Psychology from the University of Bern, Switzerland. She then pursued a PhD in Neuroscience in the Neural Control of Movement Lab at ETH Zurich, Switzerland, where she investigated the brain's ability to activate and modulate sensorimotor representations in the absence of overt movements and somatosensory input. Using fMRI and TMS, she studied neuroplasticity in the sensorimotor system induced by neurofeedback training. In 2025, she joined the School of Psychology at Queen's University Belfast, UK, as a postdoctoral researcher, where she continues to investigate neurofeedback training interventions aimed at improving sensorimotor function.

Lecture Title: Characterising Sensorimotor Representations with Multivariate Pattern Analysis

Date & Time: 17th August 2026, 9:00am BST
Meeting Link: https://teams.microsoft.com/meet/392837653750929?p=i1AVw3V57oHNjeKJuz

Dr. Saugat Bhattacharyya

Bio: Saugat Bhattacharyya is a Lecturer in Computer Science in the School of Computing, Engineering Intelligent Systems. His research interests are in the area of Brain-Computer Interfacing, Neurotechnology, Human Cognitive Augmentation, Artificial Intelligence, Data Analytics and Machine Learning and its application in Human-Machine Interaction and Neuro-Rehabilitation. His research is primarily focused on developing brain-computer interfacing systems based on robust signal processing, quantitative and machine learning algorithms to draw inference into an users' state of mind through their neural and other physiological signals. He has over 70 publications in form of peer-reviewed journals and international conferences. He is a recipient of funding from US-Ireland RD Partnership Programme, MRC Equipment Grant and GCRF pump-priming as co-investigator and two PhD fellowships by CSIR, India and Erasmus Mundus. He is currently serving as treasurer in the IEEE-EMBS UK-Ireland chapter and as associate editor/section board member in Nature Scientific Reports, Frontiers in Medical Technology and MDPI Brain Sciences.

Lecture Title: Brain-computer Interfacing Technology to improve User Engagement and Decision Making

Date & Time: 17th August 2026, 10:45am BST
Meeting Link: https://teams.microsoft.com/meet/337162603025760?p=ZYSUmSxbUHFoenuaD2

Prof. Damien Coyle

Bio: Prof. Damien Coyle is a Professor of Neurotechnology and Director of the Bath Institute for the Augmented Human, University of Bath. He is a UKRI Turing AI Acceleration Fellow 2021-25, and was previously director of the Intelligent Systems Research Centre, Ulster University (2017-2022). His research focuses on developing AI to address challenges associated with translating electrophysiological signals into control signals in brain-computer interfaces (BCI) to enable movement-independent communication/interaction targeting assistive and augmentative communication devices, cognitive and physical rehabilitation technology, and human augmentation. He has won several prestigious international awards including the 2008 IEEE Computational Intelligence Society (CIS) Outstanding Doctoral Dissertation Award, the 2011 International Neural Network Society (INNS) Young Investigator of the Year Award and the IET and E&T Innovation of the Year Award 2018. He was an Ulster University Distinguished Research Fellow in 2011, a Royal Academy of Engineering/The Leverhulme Trust Senior Research Fellow in 2013, a Royal Academy of Engineering Enterprise Fellow in 2016-2017 and an Ulster Senior Distinguished Research Fellow in 2021. He is a founding member of the International Brain-Computer Interface Society, an IEEE Brain Technical Community Steering Committee member, and an Advisory board member for the UK Neurotechnology Innovation Network. He is the Founder and CEO of NeuroCONCISE Ltd, an award-winning, AI-enabled, wearable neurotechnology company.

More information: https://pure.ulster.ac.uk/en/persons/damien-coyle

Lecture Title: Decoding mental imagery from electroencephalography (EEG) and applications of AI-enabled wearable neurotechnology for communication and rehabilitation

Date & Time: 17th August 2026, 3:00pm BST
Meeting Link: https://teams.microsoft.com/meet/357998933280091?p=1qvdwmum3Tdj1vccYw

Prof. Tony Prescott

Bio: Tony Prescott is Professor of Cognitive Robotics at the University of Sheffield in the UK. His background mixes psychology (MA Edinburgh), neuroethology and brain theory with robotics and AI (MSc Aberdeen, PhD Sheffield), and his research aims at answering questions about natural intelligence by creating synthetic entities with capacities such as perception, memory, emotion and sense of self. He has co-founded the International Living Machines conference series, and two UK companies developing robotic platforms and software. His popular science book The Psychology of Artificial Intelligence, published in 2024, explores the similarities and differences between human and artificial intelligence. His research has been covered by major news and scientific media including the BBC, CNN, Discovery Channel, Science Magazine, France 24 and New Scientist.

Lecture Title: Brain-inspired Control Architectures for Robotics and AI

Date & Time: 18th August 2026, 9:30am BST
Meeting Link: https://teams.microsoft.com/meet/385839068110527?p=5cEGn54krqKeN315pn

Prof. Elisa Donati

Bio: Dr Elisa Donati holds a B.Sc. and M.Sc. in biomedical engineering from the University of Pisa and a Ph.D. in biorobotics from Sant'Anna School of Advanced Studies. She is currently an Adj. Professor at the Institute of Neuroinformatics (INI), University of Zurich and ETH Zurich, where she also leads the research group Neuromorphic Closed-loop Systems. Elisa is part of the Neuroscience Center of Zurich.

Her work sits at the crossroads of neuroscience, neuromorphic engineering, and biomedical applications, with a focus on developing closed-loop neural interfaces, neuromorphic sensors and algorithms for biomedical signal processing, and spiking neural networks for real-time sensory encoding. She is partner in several international and national projects funded by the Royal Society UK, SNSF, and NATO.

Lecture Title: Neuromorphic Systems

Date & Time: 18th August 2026, 11:15am BST
Meeting Link: https://teams.microsoft.com/meet/35365913805120?p=pKJHGoj0ZLrUBr6yZX

Dr. Judith Evers

Bio: Dr. Judith Evers is an Assistant Professor (Ad Astra Fellow) at UCD School of Veterinary Medicine leading a small research group on preclinical neuromodulation. She holds a degree in veterinary medicine from Leipzig University, Germany, and worked in the equine sector for 2 years. She completed her PhD in translational medicine at UCD School of Medicine on sacral neuromodulation in the treatment of faecal incontinence and implemented chronic sacral neuromodulation in rodents. Judith then joined the Neuromuscular Systems Lab led by Prof. Madeleine Lowery at UCD School of Electrical and Electronic Engineering as a postdoctoral researcher, where she characterised the electrode-tissue interface of chronically implanted cerebral electrodes with the SfI CURAM Centre for medical device development which affects efficacy, implemented closed-loop deep brain stimulation in rodents where the applied stimulation adapts to current symptom severity and studied directional deep brain stimulation leads in a Boston Scientific and SfI Insight Centre co-funded project. Her research team currently focussed on novel stimulation algorithms for brain stimulation and new stimulation location in the treatment of Parkinson's disease.

Lecture Title: Introduction to neuromodulation – using translational models to understand and improve neuromodulation therapies

Date & Time: 18th August 2026, 2:15pm BST
Meeting Link: https://teams.microsoft.com/meet/339412148940641?p=Qm3WEf2yoxk5iDrFxL

Profiles of Tutors

Toby Newey

Bio: Toby Newey is a PhD researcher at the Intelligent Systems Research Centre in Ulster University, researching the augmentation of group decision-making using passive, collaborative brain-computer interfaces and computational modelling. He completed his MEng in Biomedical Engineering at Nottingham Trent University in 2024, writing his Master's thesis on the development of a multi-modal (fNIRS/EEG) neurofeedback system. Outside of academia, Toby is learning Portuguese and enjoys walking and foraging wild mushrooms.

Mark Butler

Bio: Mark Butler is a PhD researcher at the Intelligent Systems Research Centre, Ulster University. He completed his Computer Science undergraduate degree at Ulster University, Magee. His current research focuses on Brain-Computer Interface (BCI) systems, particularly using MEG and EEG modalities. He is especially interested in the role of boredom within neurorehabilitation contexts and how cognitive states can be monitored and adapted to in real time. His broader research interests include cognitive neuroscience, passive BCI design, and human-computer interaction.

Ava Rijkers

Bio: Ava Rijkers is a PhD researcher at the Intelligent Systems Research Centre in Ulster University. She completed a BSc in Computer Science at Ulster University in 2025. Ava's current research focuses on developing a learning rule that uses astrocyte Ca^2 waves as a sampling clock that gates synaptic learning in a spiking neural network.

Emmet Thompson

Bio: Emmet Thompson graduated from Ulster University in June 2025 with a BSc in Computer Science. His PhD research area combines computational neuroscience and artificial intelligence. He aims to study how interactions between neurons and astrocytes can affect learning processes in the brain. For he research he will develop computational models with neurons and astrocytes, and aim to extract biological processes that could be useful for the creation of more effective and efficient AI. Emmet is currently looking at how astrocytic calcium dynamics can affect dendritic plateau potentials. By modelling this he hopes to create an interpretable, biologically grounded learning rule based on the interactions seen.

Brendan Lenfesty

Bio: Brendan Lenfesty received his B.Sc. in Computer Science in 2021 from Ulster University, Magee campus. He completed his Ph.D. in March 2026 on Computational Modelling and Machine Learning in Decision Neuroscience at Ulster University's Magee campus. His research focuses on using computational modelling and machine learning to gain further knowledge of abstract decisions and the mechanistic processes that underly perceptual decisions.

City of Derry ∼ Londonderry in Northern Ireland

Located in the Northwest of Ireland where The Wild Atlantic Way meets the Causeway Coastal Route, the vibrant city of Derry ∼ Londonderry is renowned for one of the finest Walled Cities in Europe and home to award winning museums, some of the islands best cultural attractions and a variety of lively festivals and events; Derry ∼ Londonderry offers a vibrant social scene where your visitors are guaranteed the warmest of welcomes and hospitality. For delegates looking to experience the local culture, the city walls surround cosy pubs with live music, award-winning museums that tell stories from times past, and vibrant eateries that serve up LegenDerry Food.

This is a special wee place like no other. Our unique geography and diverse climate create the ideal conditions for our food and drink industry to flourish. Our produce harvested and crafted locally from both 'land and lough' is influenced by the latest food trends worldwide. The shores of Lough Foyle provide a vast array of shellfish with the Lough Foyle Irish Flat Oyster being the jewel in the Foyle's crown.

There's so much to discover in the Walled City with bucket loads of activities to suit all tastes. Derry is home to it all! Discover our 400-year-old City Walls, award-winning museums and theatres or why not try your hand at one of our water attractions, like Stand-Up Paddle-boarding. Take a step through history and go on a walking tour – we promise they won't disappoint. Or perhaps you would like to discover all things Derry Girls – no problem. If it's a Derry Girls themed afternoon tea, or screen walking tour you're after then we've got that on offer too. There really is something for everyone in the city; be inspired by the options below or build your own itinerary from our planner. Don't forget to buy our Visit Derry pass which means you can explore the city and enjoy access to several of the city's top tourist attractions.

Further information:

- Visit Derry (https://www.visitderry.com/)
- Discover Northern Ireland (https://discovernorthernireland.com/information/product-catch-all/visit-derry-information-centre-p689591)
- Derry City and Strabane (https://www.derrystrabane.com/services/tourism/visitor-information-services)

On campus eateries (opened during daytime):

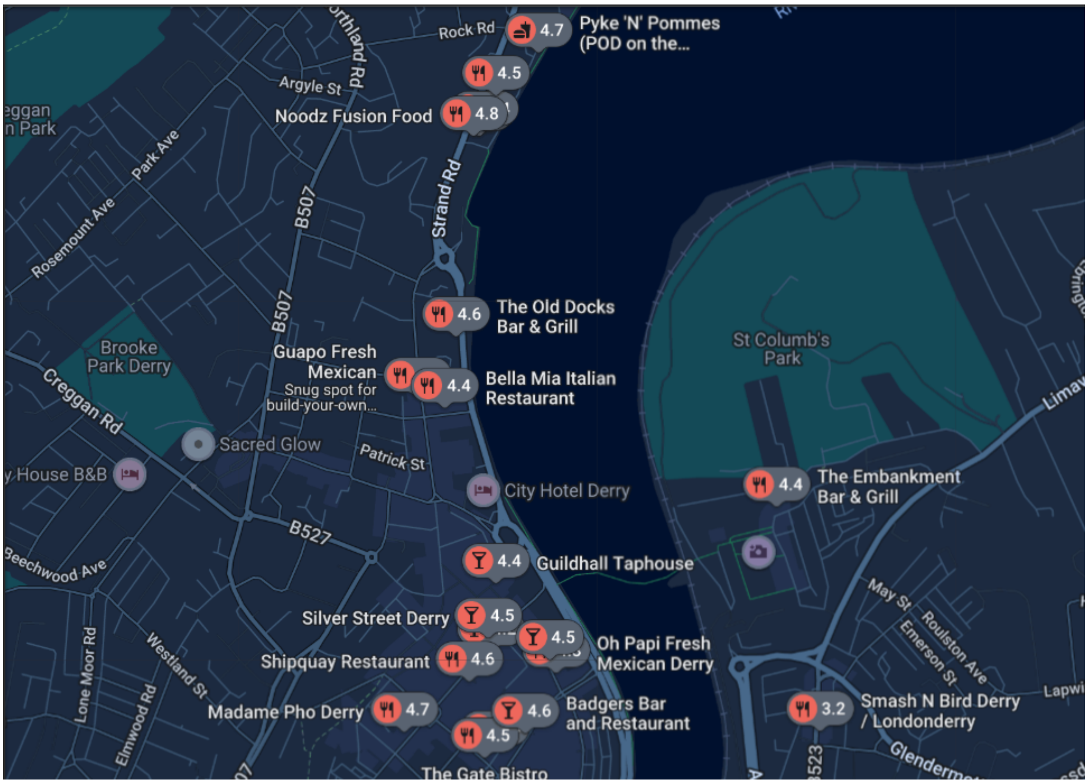
- Jitters (in MG building)
- Rock Road Social (in MF building – entrance at the back)
- Scullery Magee (in MU building)

Nearby restaurants and eateries:

- Guapo (Fresh Mexican)
- Clipper Quay Street Food Market
- Pyke 'N' Pommes (in a bus, along the Foyle River; or one along Strand Road)
- Florentini
- Quaywest
- Mama Masala
- BURN on the river (along Foyle River)
- Saffron Modern Indian Takeaway
- Mandarin Palace
- The Old Docks Bar & Grill
- Madame Pho
- Browns in Town/Bonds Hill
- Pickled Duck
- Fitzroys
- Castle Bistro (Craft Village)
- Patricia's Coffee House (along Foyle River)
- Shipquay Restaurant
- Zora's
- Sandy's African Food Hub

- Domino's Pizza (cityside)
- Mekong Street Food
- El Tapas Gra
- Walled City Brewery
- La Sosta Restaurant

etc.



Bars and pubs:

- Paedar O'Connell's
- Blackbird
- Sandino's Café Bar
- Guildhall Taphouse
- Bennigans Bar
- Grand Central Bar
- The Gweedore Bar
- Kaboodle Bar

etc.

Ulster University, Derry ∼ Londonderry (Magee campus)

The Magee campus of Ulster University in the city of Derry ∼ Londonderry, is one of four campuses in Northern Ireland: https://www.ulster.ac.uk/campuses/magee

It is the oldest campus with a history, dating back to the year 1865.

Derry Londonderry campus map:

- https://www.ulster.ac.uk/maps/derry-londonderry
- Google Map: https://www.google.com/maps/d/viewer?mid=1gdsugbd1SrO_vMTlhmyxvojrR-I&ie=UTF8&t=h&oe=UTF8&msa=0&ll=55.002408815163822C-7.3218749999999995&z=16

How to get to Derry ∼ Londonderry campus?

The MS building at Magee campus lies on Strand Road opposite the Derry City and Strabane District Council.

By Air: The City of Derry airport (https://www.cityofderryairport.com/) is the nearest airport. Or you may fly to Belfast International Airport (https://belfastairport.com/), the next closest airport, or George Best Belfast City Airport (https://www.belfastcityairport.com/). The City of Derry airport (https://www.cityofderryairport.com/destinations/) is only 7 miles from Derry ∼ Londonderry city centre. Direct flights from London Stansted, Manchester, Liverpool, Glasgow and Edinburgh. From Belfast International Airport (https://belfastairport.com/) or George Best Belfast City Airport (https://www.belfastcityairport.com/), it is 1 hour 15 minutes and 1 hour 30 minutes from Derry ∼ Londonderry city centre (see coach below), respectively. From Dublin airport (https://www.dublinairport.com), it is 2 hours 45 minutes to Derry ∼ Londonderry by car or bus.

By rail: Take the Translink (https://www.translink.co.uk/) and stop at Derry ∼ Londonderry train station. For example, from Belfast Great Victoria train station, to Derry ∼ Londonderry train station, it takes about 2 hours. To go from Dublin (Dublin City, Connolly Rail Station) to Derry ∼ Londonderry train station, you have to change trains at Belfast Lanyon Place (formerly Belfast Central) train station. There is (some) wifi service on the trains but no food service. It is better to consume or takeaway food at a train station.

By Bus: There are many buses. For example, bus 212 takes you from Belfast's Europa bus station (besides Belfast Great Victoria Station) to Derry ∼ Londonderry bus station in about 1.5 hours. There are also buses (Dublin Coach Services https://www.translink.co.uk/usingtranslink/specialoffers/dublincoachwebsaver) straight from Dublin Airport to Derry ∼ Londonderry bus station and back or from Dublin Busáras Bus (https://www.buseireann.ie/) to Derry ∼ Londonderry bus station. This is about 4 hours of journey with a break halfway. There is also an economic coach (Aircoach https://www.aircoach.ie/) from Dublin Airport straight to Belfast city, near the Belfast Great Victoria train station and Europa bus station (see above). Another option is the Aircoach service from Belfast International Airport (BFS) to Derry ∼ Londonderry Foyleside coach park.

Car hire and taxi service available from airports.

By taxi: Ask the taxi driver to stop at The Gatelodge, which is besides the MS building.

Driving from Strand Road/from Quayside roundabout

After turning in from Strand Road, please slow down and take a first turn on the left after the roundabout and after the traffic lights.

Driving from Foyle Bridge: Pass the Derry City and Strabane District Council, then do a U-turn at a roundabout and slow down and take the first turn on the left right after the traffic lights.

On campus parking:

To park at the ISRC / MS building (parking space P5 – see campus map), collect a parking ticket and use an available parking space underneath the MS building.

Go back to the front entrance, please press the disabled door opener and register at the reception someone. Please take a seat and one of our team members will be with you shortly.

There are also other parking spaces. The largest on campus parking space is P1 facing the neo-gothic-looking MD building.

electric car charger point:

For anyone needing a charging point for an electric car, you can find one in the main car park just below the MD building.

Off campus parking:

To park outside the campus, nearby parking spaces include the Strand Road Car Park, Quayside Shopping Centre & Car Park, and Foyle Street Car Park. However, for the evening lab sessions, it is advisable to park on campus. For instance, if you happen to park outside campus e.g. due to lack of available on-campus parking space, then during dinner break, for convenience, you may wish to move your car and park on campus when it becomes less crowded.

Ulster Visitor Wi-Fi:

Ulster University offers visitor Wi-Fi, which can be accessed by registering on the sign-in form with your name and email address. For more information, please visit the UU Visitor webpage. (https://www.ulster.ac.uk/ds/services/wifi-services/visitor-wifi)

Intelligent Systems Research Centre & Summer School

Address of our Research Centre:

Intelligent Systems Research Centre,

School of Computing, Engineering and Intelligent Systems,

Faculty of Computing, Engineering and the Built Environment,

Ulster University,

Magee campus

Northland Road,

Derry ∼ Londonderry,

BT48 7JL,

Northern Ireland, UK

Note: The Intelligent Systems Research Centre is also the MS building on Magee campus.

Sponsors